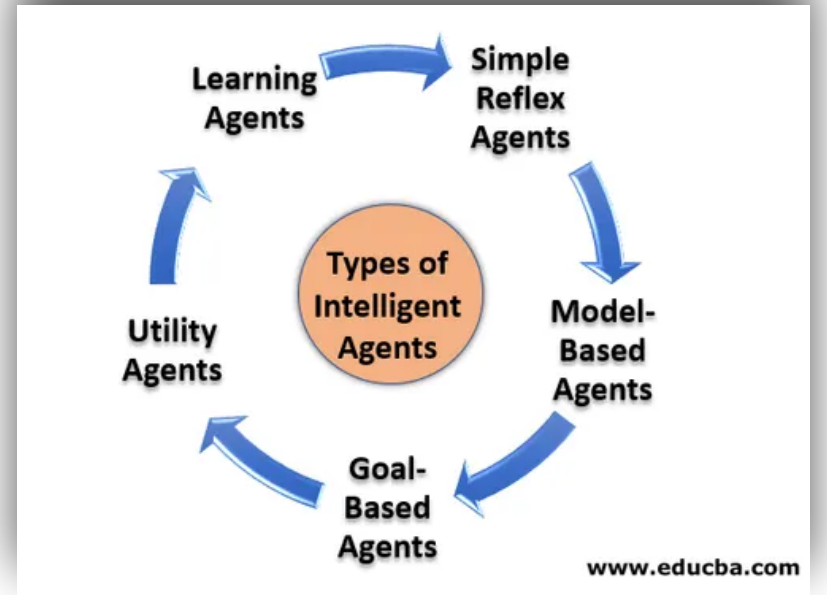
CSE276 — AI FOUNDATIONS | WEEK 7 | LECTURE 13 | UNIT III

Agent Architectures

How intelligent agents decide what to do



The Question From Last Lecture

We know HOW an agent perceives and acts.

But what happens **inside** the agent?



Core Question

If an agent receives information from the environment, HOW does it decide what action to take?

This is defined by its Agent Architecture.

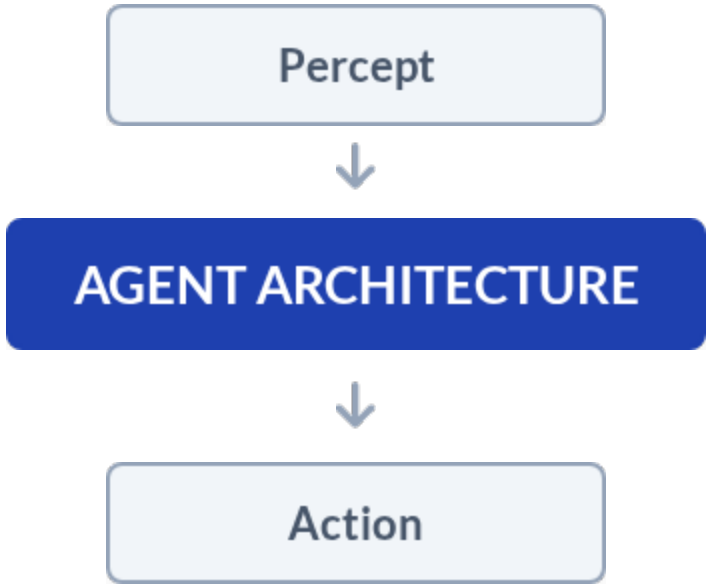
Learning Objectives

By the end of this lecture, you should be able to:

- Explain what an agent architecture is.
- Explain how a simple reflex agent selects actions.
- Understand the limitation of simple reflex agents.
- Explain how model-based agents maintain internal state.
- Explain goal-based decision-making.
- Explain why utility provides a richer decision criterion.
- Explain the basic idea of learning agents.
- Compare the five architectures.
- Identify suitable architectures for simple AI problems.
- Connect the architectures to Practical 5.

What Is an Agent Architecture?

> An **agent architecture** describes how an agent uses what it perceives to decide what action to take.



The architecture determines how the agent:

- Uses current information
- Uses previous information
- Compares outcomes
- Learns from experience

The Five Architectures as an Evolution

Each architecture adds more capability for dealing with complex decision-making.



React **Remember** **Plan** **Evaluate** **Improve**

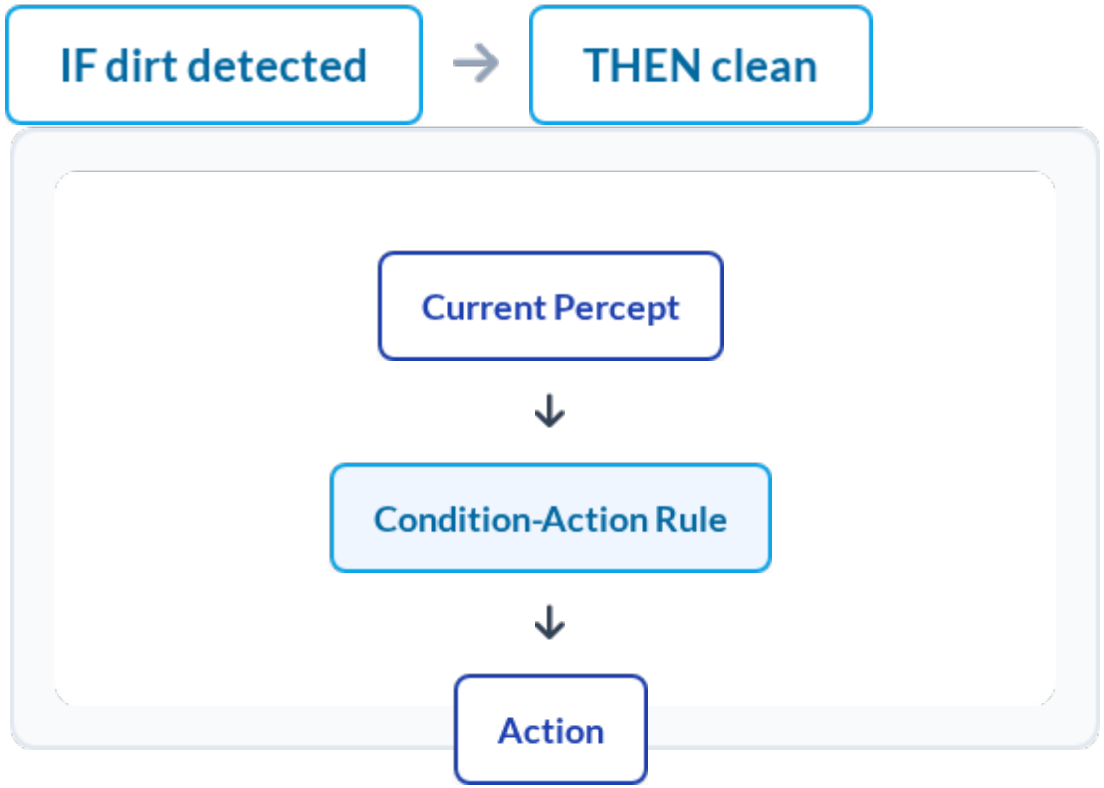
**Note: Do not imply that every learning agent must contain all five as rigid sequential stages. This is a conceptual progression of capabilities.*

1. Simple Reflex Agent

> A simple reflex agent selects actions based **only on the current perception**.



Rule Example: Robot Vacuum



Simple Reflex Agent: Examples

These systems respond directly to what they currently perceive.



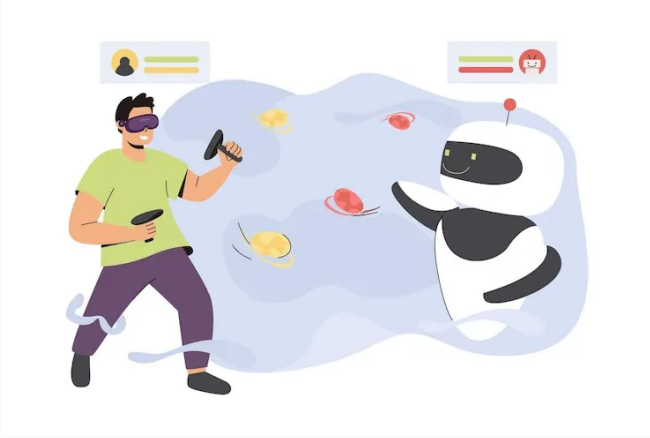
Robot Vacuum

```
IF dirt → clean
```



Smart Thermostat

```
IF temp < target → heat
```



Game AI Enemy

```
IF attacked → defend
```

What happens if the agent cannot see what is happening?

Limitation of Simple Reflex Agents

Scenario

A robot vacuum enters a room. It cannot see behind the furniture.

It bumps into an obstacle and turns away.



Question

Does it know that it has already visited this area?

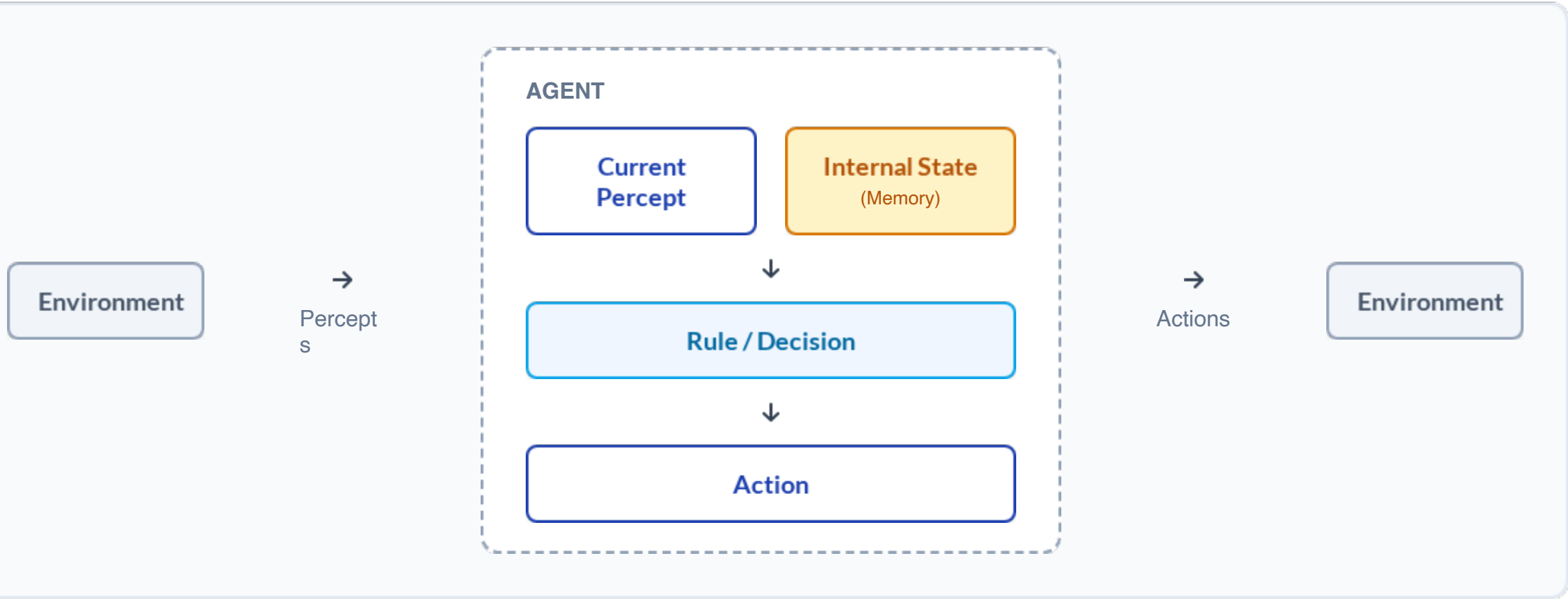
No.

A simple reflex agent **does not maintain a useful internal memory** of the world. It only knows what it sees *right now*.

Transition: Maybe the agent needs some memory.

2. Model-Based Reflex Agent

A model-based agent maintains an **internal state** that helps it keep track of aspects of the world that cannot be observed directly.



Why Do We Need an Internal State?

Let's look at the robot vacuum example again.

Time 1

Robot is cleaning **Room A**.

Perceives Room A

Time 2

Robot moves to **Room B**.

Even if it cannot see Room A anymore, it maintains an internal state:

"I already cleaned Room A."

Current Percept + Memory = Better understanding of situation

The internal state is the agent's maintained information about the world.

Simple Reflex vs Model-Based

Feature	Simple Reflex	Model-Based
Decision flow	Current percept → Rule → Action	Current percept + State → Decision → Action
Uses current percept?	Yes ✔	Yes ✔
Maintains internal state?	No ✘	Yes ✔
Memory of past info?	No ✘	Yes ✔
Handles hidden info?	Limited / Poorly	**Better**

Which architecture is better suited to a partially observable environment?

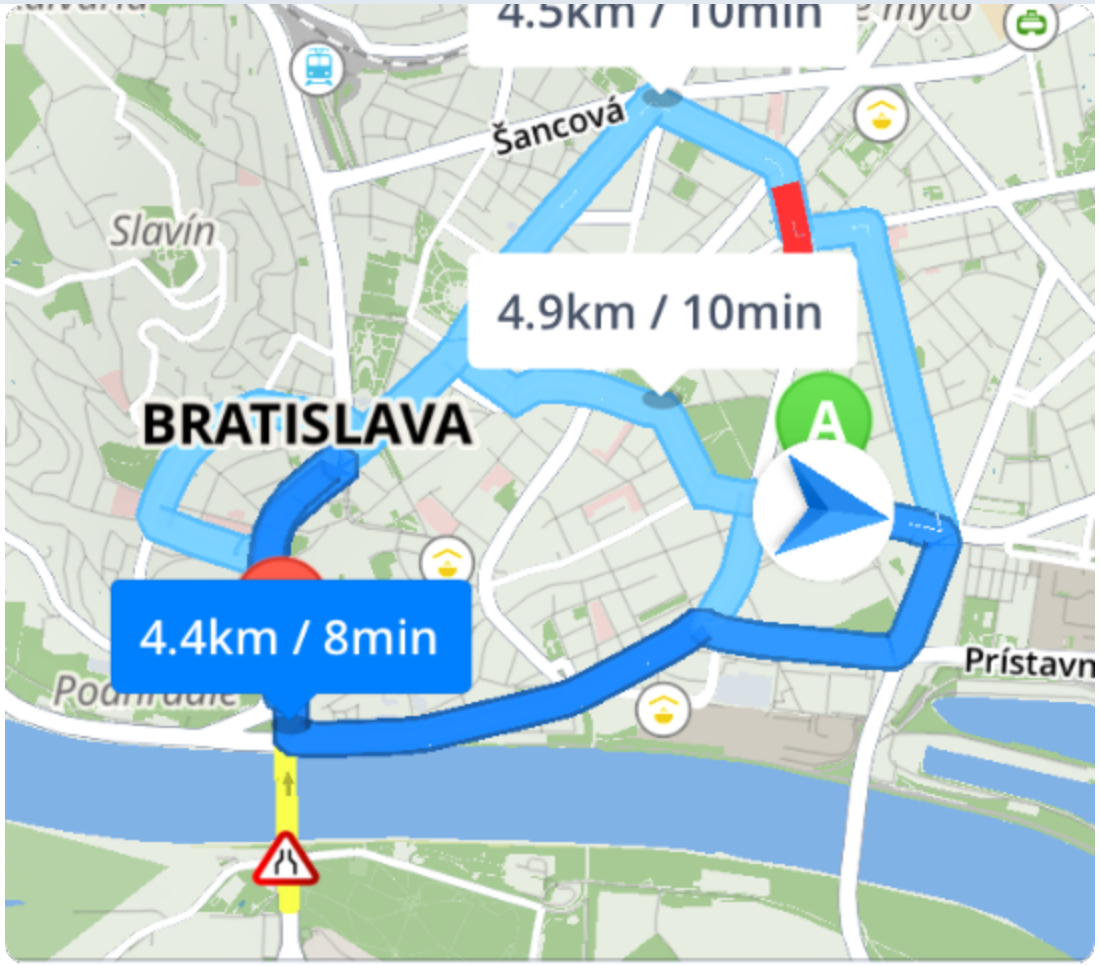
Model-based.

Why Rules Aren't Always Enough

Scenario

A navigation system is at a junction.

> **Possible actions:**
>
> Turn left
>
> Turn right
>
> ↑ Continue straight



A simple rule cannot always tell us which action is best.
Why?

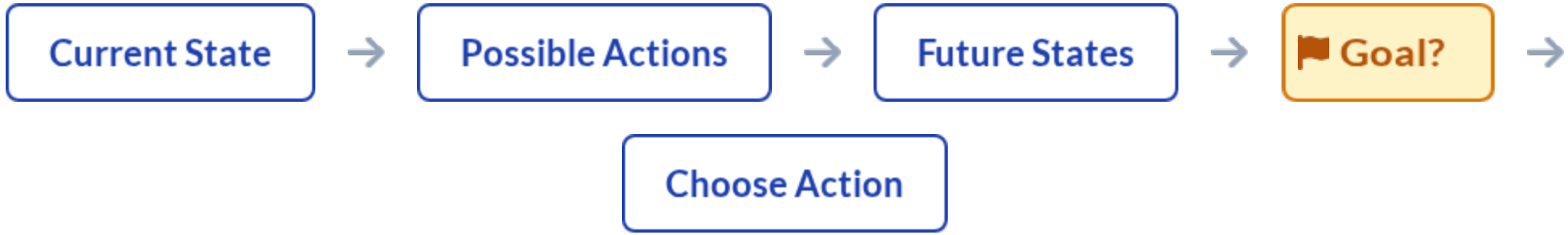
Because the agent needs to know:

> **"What are we trying to achieve?"**

Introduce: GOALS.

3. Goal-Based Agent

A goal-based agent selects actions by considering whether they help achieve a **desired goal**.



Example: Navigation Agent

Goal: Reach the destination.

The agent considers possible routes and future states rather than simply reacting to the current percept.

Goal-Based Example: Chess

Goal: Win the game

Possible actions right now:

- Move knight
- Move bishop
- Move pawn
- Castle

A goal-based agent considers the **possible consequences** of these actions in relation to its goal of winning.



Correct starting position: white square on bottom-right, pieces placed properly

What If Multiple Actions Achieve the Goal?

A navigation agent has two routes to reach the destination.

Route A

- 30 minutes
- High traffic
- Bumpy road

Route B

- 35 minutes
- Low traffic
- Safer & Smoother

Both routes successfully achieve the goal.

Which one should the agent choose?

Goal alone may not be enough. Introduce: UTILITY.

4. Utility-Based Agent

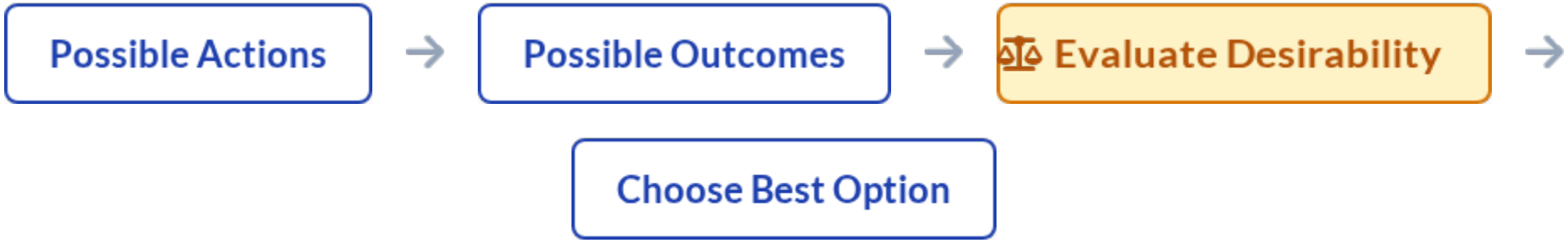
A utility-based agent evaluates how **desirable** different outcomes are and chooses an action expected to provide **higher utility**.

Goal-based asks:

"Will this help me achieve the goal?"

Utility-based asks:

"Which successful outcome is better**?"**



Utility Example

Goal: Reach destination

Route	Time	Safety	Comfort
Route A	30 min	Medium	Low
Route B	35 min	High	High

A utility-based agent can consider **multiple factors** rather than only whether the destination is eventually reached.

> **Utility provides a way to represent preferences among outcomes.**

What If the Agent Doesn't Know the Best Rules?

What if we cannot manually write every rule the agent needs?

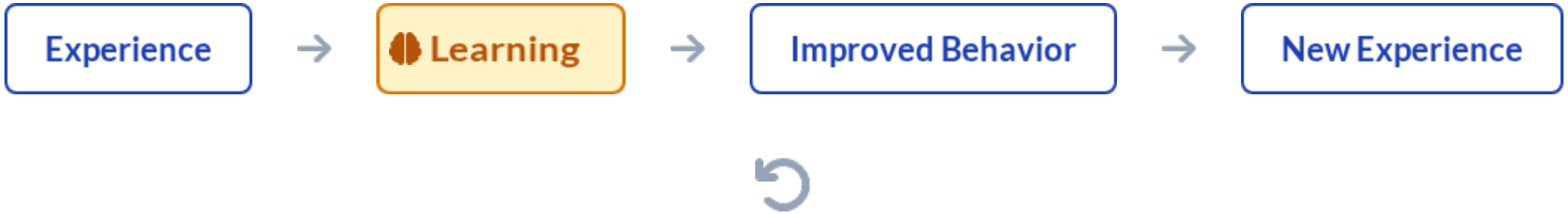
Some environments are too complex to pre-program everything:

- Recommendation systems
- Game-playing systems
- Speech recognition systems
- Advanced robot navigation

Introduce: LEARNING AGENTS

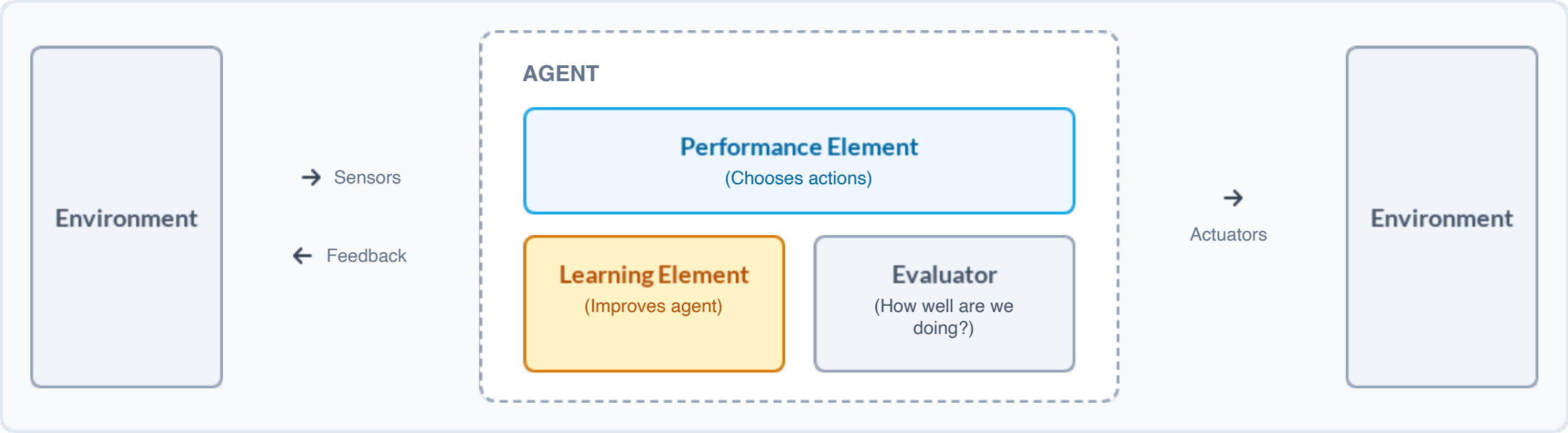
5. Learning Agent

A learning agent **improves its behavior** by using experience or feedback from its environment.



The agent does not have to be completely programmed with every useful behavior beforehand.

Basic Learning Agent Structure



Performance element: Chooses actions (like the previous architectures).

Learning element: Makes changes to improve future performance.

Feedback / Evaluator: Helps the agent determine how well it is doing.

Example: A Game-Playing AI



Learning is about improving behavior from experience.

Five Architectures Compared

Architecture	Main Idea	Memory?	Goals?	Preferences?	Learns?
Simple Reflex	Current percept → rule	No	No	No	No
Model-Based	Percept + internal state	**Yes**	Not necessarily	No	No
Goal-Based	Choose actions toward goal	Usually	**Yes**	Limited	No
Utility-Based	Choose preferred outcome	Usually	**Yes**	**Yes**	No
Learning	Improve from experience	Often	Can	Can	**Yes**

**This is a conceptual comparison, not a strict rule that every implementation must fit perfectly into one box.*

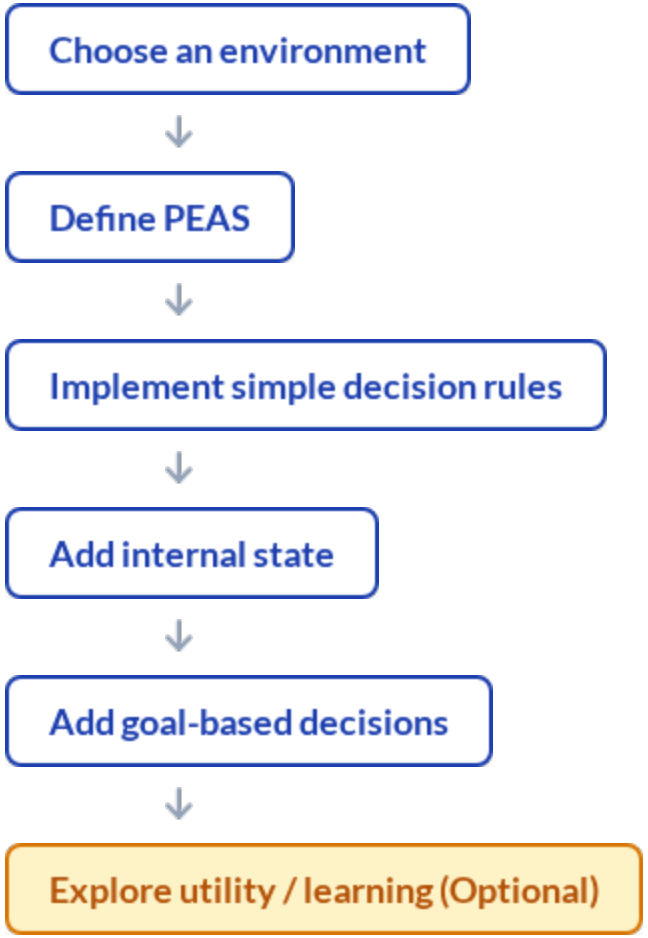
Same Robot, Five Architectures



Simple Reflex	"If dirt → clean."
Model-Based	"I remember where I've already cleaned."
Goal-Based	"I need to clean the entire room."
Utility-Based	"I'll balance cleaning, battery, and time."
Learning	**"I learn from previous cleaning experiences."**

Practical 5 — Building Agent Architectures

Suggested Workflow



Questions for Comparison

- What information does the agent use?
- Does it remember?
- Does it have a goal?
- Does it compare outcomes?
- Does it learn?

Final Recap + Exit Question

React → Remember → Plan → Evaluate → Learn

Simple Reflex | Model-Based | Goal-Based | Utility-Based | Learning

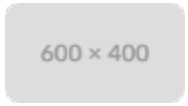
Exit Question

A self-driving car must reach a destination, avoid accidents, minimize travel time, and adapt to driving conditions based on experience.

Which agent architecture or combination of capabilities would be appropriate, and why?

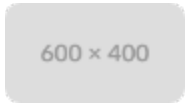
Next lecture: Multi-agent systems / agent interaction

Image Sources



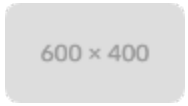
https://img.freepik.com/free-vector/cyborg-concept-illustration_114360-14988.jpg?size=338&ext=jpg&ga=GA1.1.2008272138.1724025600&semt=ais_hybrid

Source: www.freepik.com



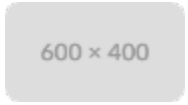
https://static.vecteezy.com/system/resources/previews/060/534/487/non_2x/robot-vacuum-cleaning-in-modern-living-room-with-minimal-furniture-showcasing-smart-home-convenience-and-efficiency-in-minimalist-interior-design-aesthetic-photo.jpeg

Source: www.vecteezy.com



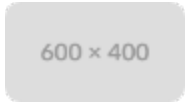
https://thumbs.dreamstime.com/b/smart-wall-thermostat-showing-temperature-d-rendering-smart-wall-thermostat-showing-temperature-d-128227092.jpg

Source: www.dreamstime.com



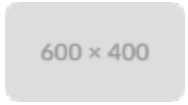
https://cdn.dribbble.com/users/2884218/screenshots/11566415/media/fb66453d10deccab8c28258dc78c2e39.png?resize=400x300&vertical=center

Source: dribbble.com



https://media.istockphoto.com/id/1160641399/photo/top-view-on-wooden-chess-board-with-figures-during-the-game-on-white-wooden-table-background.jpg?s=612x612&w=0&k=20&c=_JP5QDhfFyN7bJsLtIyZyrrlnHiIrz7E3QT1_0RN9_o=

Source: www.istockphoto.com



https://as2.ftcdn.net/v2/jpg/04/71/34/44/1000_F_471344402_wYm0j7WJ79X2bYq8Qj6O9l0n8d2p7J7J.jpg

Source: stock.adobe.com

Image Sources



https://media.easy-peasy.ai/f75451aa-55ed-4aee-8bc3-cc658a7000a1/839853e4-b779-4d2e-98df-74482f9a14f2_medium.webp

Source: easy-peasy.ai



https://maverickframe.com/wp-content/uploads/2026/03/robotic-vacuum-cleaner-design-dock-fireplace-interior.webp

Source: maverickframe.com



https://www.watts.eu/dfsmedia/0533dbba17714b1ab581ab07a4cbb521/656871-50060/639039057860000000

Source: www.watts.eu



https://lh7-rt.googleusercontent.com/docsz/AD_4nXdo2BKrM2U1zW1WvGlvR0fEfx-d1bEHKaxGyC-v4-dyoN_20Ir7DvuHaGkld_Yz8DpL1i0IsXuCm29La5IszD-g0Ix6E-J3wmpNUeQWOAMJyY1QNDcaQivMndL_Zv1WVU3mRmjo1w?key=iLoMmkiWuSj5vsrukOWwL_T3

Source: us.narwal.com



https://www-sygic.akamaized.net/content/13-what-is/0-route-computing/route_computing.png

Source: www.sygic.com



http://chessbazaar.com/cdn/shop/articles/setup_chessboard_b0c8ed71-475c-43c9-b7be-63d5d10fdf70.jpg?v=1781602056

Source: chessbazaar.com